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Database Foundations

2-5

Relationships

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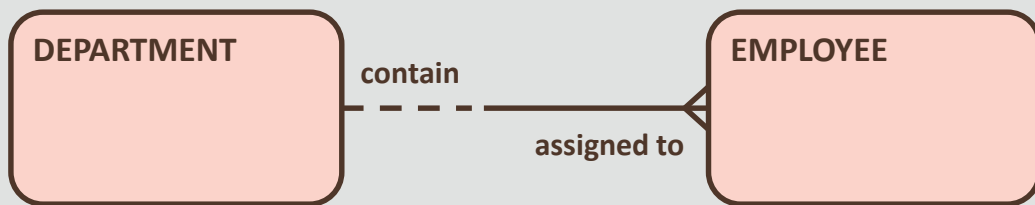
Objectives

- This lesson covers the following objectives:
 - Define and recognize examples of relationships and corresponding foreign keys
 - Identify the optionality of relationships
 - Identify the cardinality of relationships
 - Types of relationships
 - Relationship matrix



Relationships

- A relationship is a bidirectional, significant association between two entities or between an entity and itself



A relationship represents the business rules that link entities. Each relationship always has two business rules. In the slide example, the business rules are:

- A DEPARTMENT may contain one or many EMPLOYEES.
- An EMPLOYEE must be assigned to one and only one DEPARTMENT .

Relationships

- Relationships represents an association between two or more entities
- The relationship line on the diagram is either solid (mandatory) or dashed (optional)
- These lines terminate in either a "single toe" (one instance) or a "crow's foot" (one or more instance)

In a conceptual data model, a relationship is any association, linkage, or connection between the entities of interest to the business.

Relationships

- Relationships have names that help describe the connection between the entities
- In the relationship diagram the name of the relationship, from either perspective, is printed near the starting point of the relationship line (see slide 5)
- Examples:
 - DEPARTMENTS contains EMPLOYEES
 - EMPLOYEES assigned to DEPARTMENTS

Foreign Key

- Relationships in a conceptual data model are mapped to foreign keys in a physical database table
- A foreign key (FK) is a column or a combination of columns in one table that refers to a primary key in the same table or another table

Foreign Key Examples

EMPLOYEES

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
100	Steven	King	90
101	Neena	Kochhar	90
102	Lex	De Haan	90
103	Alexander	Hunold	60
104	Bruce	Ernst	60

Foreign
Key

refers to

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME
10	Administration
20	Marketing
50	Shipping

Primary Key →

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DFo 2-5
Relationships

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To link these two tables and determine employee and department information, a corresponding DEPARTMENT_ID column must be inserted in the EMPLOYEES table to reference existing department IDs in the DEPARTMENTS table. In this case, the DEPARTMENT_ID column in the EMPLOYEES table is a foreign key that references a column that has the same name in the DEPARTMENT table. This is an example of a relationship between two tables.

Components of a Relationship

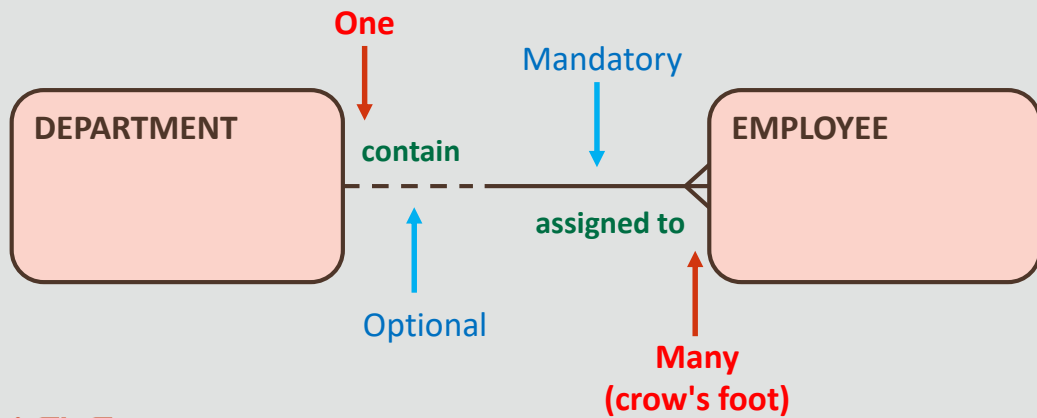
- The components of the relationship include the following:
 - Name: The label that appears close to the entity it is assigned to. Make sure that all relationship names are in lowercase
 - Cardinality: The minimum and maximum number of values in the relationship
 - One and only one matching record
 - One or more matching records
 - Optionality: Whether the relationship must exist
 - Optional (zero matching records)
 - Mandatory (at least one matching record in each entity)

Components of a Relationship

- When reading the business rule sentence, use the following wording:
 - Optional: Use "may be" or "may."
 - Mandatory: Use "must be" or "must."
 - Line: Use "one and only one."
 - Crow's feet: Use "one or more."
- Business rule syntax is as follows:
 - Each entity1 {must be or may be} relationship name {one or more or one and only one} entity2

Components of a Relationship

- Name
- Cardinality
- Optionality



Project Exercise 1

- DFo_2_5_1_Project
 - Oracle Baseball League Store Database
 - Recognize Examples of Relationships



What Is Optionality in a Relationship?

- Relationships are either mandatory or optional
- Consider the two entities, EMPLOYEE and JOB
- Based on what you know about instances of the entities, you can determine optionality by answering two questions:
 - Must every employee have a job? In other words, is this a mandatory or an optional relationship for an employee?
 - Must every job be done by an employee? In other words, is this a mandatory or optional relationship for a job?

Project Exercise 2

- DFo_2_5_2_Project
 - Oracle Baseball League Store Database
 - Identify the Optionality of Relationships



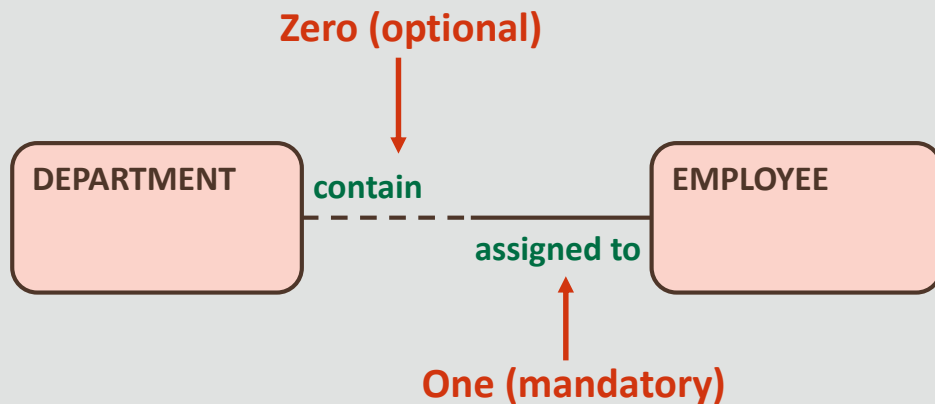
What Is Cardinality in a Relationship?

- Cardinality measures the quantity of something
- In a relationship, it determines the degree to which one entity is related to another by answering “How many?”
- For example:
 - How many jobs can one employee hold? One job only? Or more than one job?
 - How many employees can hold one specific job? One employee only? Or more than one employee?

The cardinality of a relationship answers only whether the number is singular or plural; it does not answer with a specific plural number.

Determining the Relationship's Minimum Cardinality

- What is the minimum cardinality in each direction?



The first question to answer is this: What is the minimum cardinality for each direction of the relationship?

In the slide example, answer the following questions:

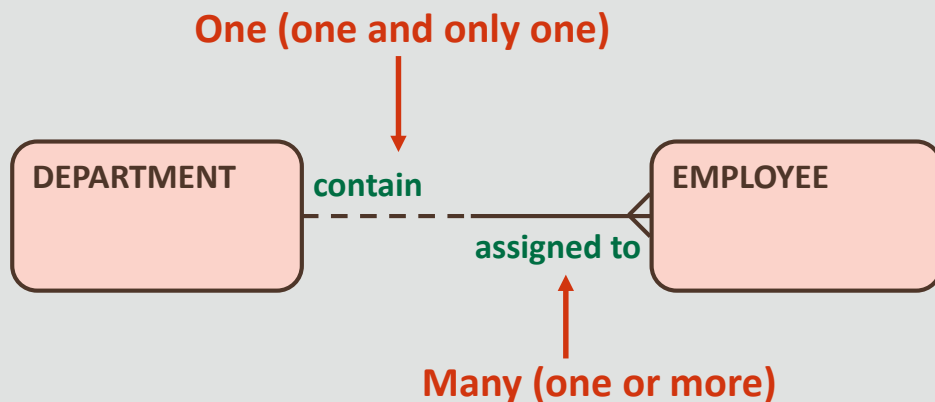
1. Must an EMPLOYEE be assigned to a DEPARTMENT? The answer is always.
2. Is there any situation in which an EMPLOYEE will not be assigned to a DEPARTMENT?
The answer is no. An EMPLOYEE must always be assigned to a DEPARTMENT. (Mandatory)
3. Must a DEPARTMENT be composed of an EMPLOYEE?
The answer is no. A DEPARTMENT does not have to be composed of an EMPLOYEE. (Optional)

When the minimum cardinality is optional, the value could be zero. When the minimum cardinality is mandatory, the value must be at least one.

Note that the relationship line in the slide was intentionally drawn without the maximum cardinality.

Determining the Relationship's Maximum Cardinality

- What is the maximum cardinality in each direction?



The second question to answer is this: What is the maximum cardinality for each direction of the relationship?

In the slide example, answer the following questions:

1. Must an EMPLOYEE be assigned to more than one DEPARTMENT?
The answer is no. An EMPLOYEE must always be assigned to one and only one DEPARTMENT. (One)
2. May a DEPARTMENT be composed of more than one EMPLOYEE?
The answer is yes. A DEPARTMENT may be composed of one or more EMPLOYEEs. (Many)

When the maximum cardinality is one, the value can be only one. When the maximum cardinality is many, the value can be one or more.

Project Exercise 3

- DFo_2_5_3_Project
 - Oracle Baseball League Store Database
 - Identify the Cardinality of Relationships

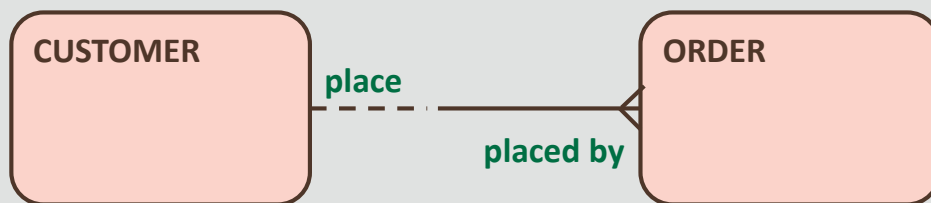


Optionality and Cardinality: Examples

- Each EMPLOYEE must hold one and only one JOB
- Each JOB may be held by one or more EMPLOYEE
- Each PRODUCT must be classified by one and only one PRODUCT TYPE
- Each PRODUCT TYPE may classify one or more PRODUCTS

Relationship Types

- All relationships represent the information requirements and the rules of the business
 - Many-to-one (M:1) or one-to-many (1:M)
 - Many-to-many (M:M)
 - One-to-one (1:1)



Example of a 1:M relationship

There are three types of relationships:

- **Many to one (M:1) or one to many (1:M):** There are crow's feet on one side of the relationship. The direction of the crow's feet determines whether the relationship is M:1 or 1:M. This type of relationship is the most common.
- **Many to many (M:M):** There are crow's feet on both sides of this relationship. It is common to see M:M relationships in a high-level ERD at the beginning of a project.
- **One to one (1:1):** This type of relationship is a line without crow's feet on either end. These types of relationships are rare.

Note: The notation is slightly different in Oracle SQL Developer Data Modeler, where one to many is 1:N and many to many is M:N.

Many-to-One and One-to-Many Relationships

- Many-to-one and one-to-many relationships (M:1 and 1:M) have cardinality of one or more in one direction and one and only one in the other direction

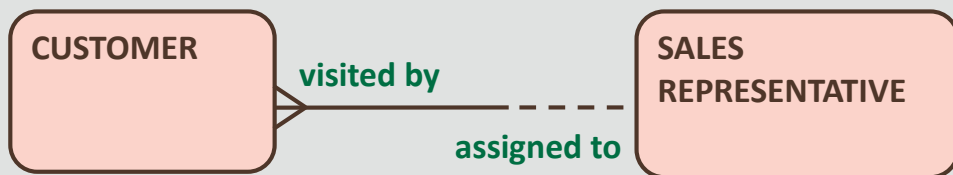


In the one- to- many (1:M) relationship, a single record in an entity (SALES REPRESENTATIVE) can be related to one or more records in another entity (CUSTOMER), but a single record in CUSTOMER can be related to only to one SALES REPRESENTATIVE.

Many-to-One and One-to-Many Relationships

- Business rules:

- Each CUSTOMER must be visited by one and only one SALES REPRESENTATIVE
- Each SALES REPRESENTATIVE may be assigned to one or more CUSTOMER



Many-to-Many Relationships

- Many-to-many relationships (M:M) have cardinality of one or more in both directions



- Business rules:
 - Each EMPLOYEE may be assigned to one or more JOB
 - Each JOB may be carried out by one or more EMPLOYEE

In the many- to- many (M:M) relationship, many records in one entity match many records in another entity.

One-to-One Relationships

- One-to-one relationships (1:1) have cardinality of one and only one in both directions



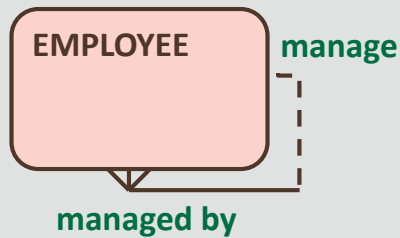
- Business rules:
 - Each **COMPUTER** must contain one and only one **MOTHERBOARD**
 - Each **MOTHERBOARD** must be contained in one and only one **COMPUTER**

The slide example is a one-to-one relationship because the cardinality is a line with no crow's feet in either direction.

These types of relations are the least common because they may instead be one entity with attributes contained in that entity.

Recursive Relationships

- A recursive relationship is a relationship with an entity and itself



- Business rules:
 - Each **EMPLOYEE** may manage one or more **EMPLOYEE**
 - Each **EMPLOYEE** must be managed by one and only one **EMPLOYEE**

Relationship Matrix: Collecting Information

- A relationship matrix can be used to collect initial information about the relationships among a set of entities

	CUSTOMER	ITEM	ORDER	WAREHOUSE
CUSTOMER			place	
ITEM			contained on	stored in
ORDER	placed by	issued for		
WAREHOUSE		store		

Relationship Matrix: Collecting Information

- A relationship matrix has the following characteristics:
 - A relationship matrix shows if and how each row entity on the left side of the matrix is related to each column entity shown across the top of the matrix
 - All entities are listed along the left side of the matrix and at the top of the matrix
 - If a row entity is related to a column entity, the name of that relationship is shown in the intersection box
 - If a row entity is not related to a column entity, the intersection box is empty
 - Each relationship above the diagonal line is the inverse or mirror image of a relationship below the line
 - Recursive relationships are represented by the boxes on the diagonal

Relationship Matrix: Mapping the Contents

- Map the contents of the relationship matrix to an ERD.

	CUSTOMER	ITEM	ORDER	WAREHOUSE
CUSTOMER			place	
ITEM			contained on	stored in
ORDER	placed by	issued for		
WAREHOUSE		store		

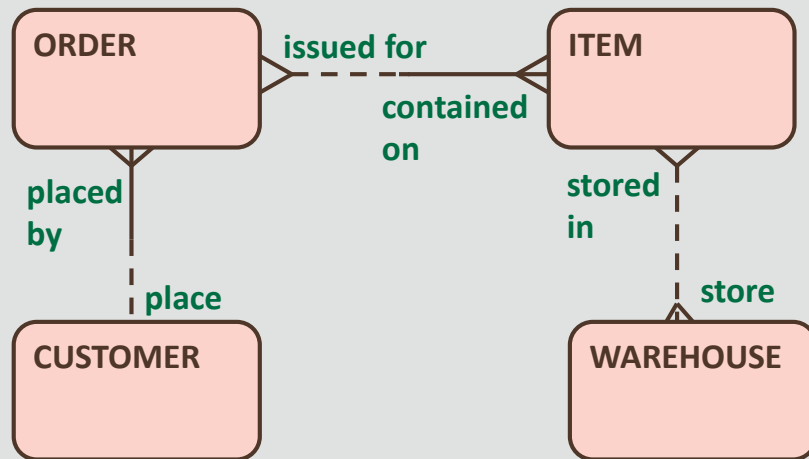
To map the matrix to an ERD, you draw a box for each entity and then draw the relationship and state the business rule. Doing so helps to determine each relationship's type and cardinality.

To help in the transformation between the matrix and the ERD, follow these steps:

- Determine a relationship's existence
- Name the relationship
- Determine the relationship's cardinality

Relationship Matrix: Mapping the Contents

- Results of mapping the contents of the relationship matrix to an ERD:



Determining a Relationship's Existence

- Examine each pair of entities to determine whether a relationship exists

	ACTIVITY	DEPARTMENT	EMPLOYEE
ACTIVITY			
DEPARTMENT			
EMPLOYEE			

The first step in creating a relationship is to determine its existence. Ask yourself, “Does a significant relationship exist between ENTITY A and ENTITY B?”

In the slide example, consider the following questions:

- Is there a significant relationship between DEPARTMENT and EMPLOYEE? The answer is yes.
- Is there a significant relationship between DEPARTMENT and ACTIVITY? The answer is no.
- Is there a significant relationship between ACTIVITY and EMPLOYEE? The answer is yes.

Log the relationships among ACTIVITY, DEPARTMENT, and EMPLOYEE on a relationship matrix. The check marks indicate that a relationship exists.

A relationship matrix is used to systematically examine each pair of entities.

Determining a Relationship's Existence

- Examine each pair of entities to determine whether a relationship exists

	ACTIVITY	DEPARTMENT	EMPLOYEE
ACTIVITY			✓
DEPARTMENT			✓
EMPLOYEE	✓	✓	

The first step in creating a relationship is to determine its existence. Ask yourself, “Does a significant relationship exist between ENTITY A and ENTITY B?”

In the slide example, consider the following questions:

- Is there a significant relationship between DEPARTMENT and EMPLOYEE? The answer is yes.
- Is there a significant relationship between DEPARTMENT and ACTIVITY? The answer is no.
- Is there a significant relationship between ACTIVITY and EMPLOYEE? The answer is yes.

Log the relationships among ACTIVITY, DEPARTMENT, and EMPLOYEE on a relationship matrix. The check marks indicate that a relationship exists.

A relationship matrix is used to systematically examine each pair of entities.

Naming the Relationship

- Name each direction of a relationship

	ACTIVITY	DEPARTMENT	EMPLOYEE
ACTIVITY			assigned to
DEPARTMENT			contain
EMPLOYEE	participate in	assigned to	

Relationship names represent a role and tend to be passive verbs, noun role names, or prepositions. Try not to use "related to" or "associated with" as relationship names because they lack specific meaning and are weak.

The following is a sample list of relationship name pairs to assist you in naming relationships:

- based on/the basis for
- bought from/the supplier of
- description of/for
- operated by/the operator for
- represented by/the representation of
- responsible for/the responsibility of

Log the relationship names in the relationship matrix as shown in the slide.**

Project Exercise 4

- DFo_2_5_4_Project
 - Oracle Baseball League Store Database
 - Using a Relationship Matrix



Summary

- In this lesson, you should have learned how to :
 - Define and recognize examples of relationships and corresponding foreign keys
 - Identify the optionality of relationships
 - Identify the cardinality of relationships
 - Types of relationships
 - Relationship matrix



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